Toward Uncertainty-Aware and Secure Multi-Agent AI Systems for High-Stakes Decision Making

1. Research Vision

My research focuses on developing reliable and safe AI systems that operate under uncertainty in high-stakes environments, particularly where multiple AI components interact as autonomous or semi-autonomous agents.

Modern AI systems are rapidly evolving from single predictive models into complex agentic systems that combine vision models, large language models, retrieval modules, and generative models in multi-step decision pipelines. While these systems demonstrate impressive capabilities, they introduce new failure modes such as compounded hallucination, hidden uncertainty propagation, and cross-agent misinformation amplification.

My long-term goal is to build uncertainty-aware, interpretable, and secure multi-agent AI systems, with a focus on understanding and mitigating emergent risks in generative and reasoning models, especially in domains like healthcare where errors are high-impact.

2. Background and Prior Work

My research background is in Biomedical Artificial Intelligence, where I have worked across diffusion models, Bayesian deep learning, and multi-agent learning systems for medical imaging.

2.1 Diffusion Models for Medical Data Generation and Completion

I have worked on diffusion-based generative models for chest X-ray synthesis and reconstruction etc. My work explores how generative models can be used not only for data augmentation but also for structured medical image completion and representation learning.

Beyond generation quality, this work raised an important question:

How do synthetic or partially generated samples influence downstream reasoning and decision systems? Which directly connects to safety concerns in generative AI pipelines, especially when synthetic data is used for training downstream models.

2.2 Bayesian Deep Learning and Uncertainty Estimation

I have explored uncertainty-aware deep learning methods, including MC Dropout and evidential learning, for medical image classification. A key focus has been measuring and improving model calibration and predictive reliability with uncertainty quantification.

However, I observed a limitation in existing approaches: uncertainty is typically estimated at the *single-model level*, without considering how uncertainty behaves in multi-stage or multi-agent pipelines. This motivates the need for system-level uncertainty modeling in agentic AI.

2.3 Multi-Agent Medical AI Systems

I have developed multi-agent frameworks for chest X-ray interpretation, where different agents perform:

- disease classification
- report generation
- uncertainty estimation

This setting revealed a critical phenomenon:

errors and overconfident predictions can propagate across agents and become mutually reinforced.

This is analogous to emerging concerns in AI safety research such as deceptive alignment, self-reinforcing hallucinations, and agent interaction failures.

2.4 LLM-Based Report Generation and Reasoning

I have also worked on LLM-based medical report generation systems, where vision-language models generate structured clinical narratives.

A key limitation observed is that:

- fluent reports can mask incorrect visual grounding
- uncertainty is often lost during natural language generation
- downstream users may over-trust generated explanations

This raises important safety concerns around interpretability, calibration, and contextual integrity in LLM outputs.

3. Research Direction Going Forward

My future research focuses on three interconnected directions:

3.1 Uncertainty-Aware Multi-Agent AI Systems

I aim to develop frameworks where uncertainty is explicitly propagated, monitored, and regulated across interacting AI agents.

Key questions include:

- How does uncertainty evolve in multi-agent reasoning pipelines?
- Can agents detect when they are reinforcing incorrect or overconfident beliefs?
- How can uncertainty signals be used as a control mechanism for safer decision-making?

This extends Bayesian deep learning into system-level safety modeling for agentic AI.

3.2 Safety and Failure Modes in Generative and LLM-Based Systems

Generative models and LLMs introduce unique risks:

- hallucinated reasoning chains
- misaligned natural language explanations
- over-trust in fluent outputs

I am interested in studying how diffusion models + LLM pipelines behave under adversarial or ambiguous conditions, and how failures propagate when generation is part of a decision system.

3.3 Evaluation Frameworks for Agentic Medical AI Systems

A major gap in current research is the lack of system-level evaluation metrics for multi-agent AI.

I aim to develop evaluation protocols that measure:

- uncertainty calibration across agents
- propagation of incorrect reasoning
- robustness under distribution shift and adversarial inputs
- consistency between generated reports and visual evidence

4. Fit with AI Safety and COMPASS Vision

My research directly aligns with the COMPASS group's focus on:

- secure and cooperative multi-agent systems
- contextual integrity in AI reasoning
- evaluation of agentic risks
- interpretability and control of LLM-based systems
- failure modes such as deception, hallucination, and misalignment

My background in diffusion models, Bayesian uncertainty, and LLM-based medical reasoning systems provides a technical foundation for studying how such risks emerge in real-world multimodal AI systems, particularly in high-stakes domains like healthcare.

5. Conclusion

I aim to contribute toward building safe, interpretable, and uncertainty-aware AI systems that remain reliable under complex multi-agent interactions and generative pipelines. By combining ideas from Bayesian deep learning, diffusion models, and LLM-based reasoning systems, my goal is to help develop principled methods for evaluating and controlling the behavior of next-generation agentic AI systems, ensuring they remain aligned, calibrated, and robust in real-world deployments.